

CONDITIONAL STATEMENTS

Conditional Statements



All **Conditional Statements** in Excel are based on simple “IF/THEN” statements:

- IF* it's raining, **THEN** bring an umbrella
- IF* it's sunny, **THEN** bring sunglasses
- IF* it's sunny **AND** it's summer, skip work and go to the beach

You're basically saying ***“Hey Excel, if this statement is true, do this. Otherwise, do something else.”***

=IF(logical_test, [Value if True], [Value if False])

Any test that results in either **TRUE** or **FALSE**

(i.e. A1="Google", B2<100, etc)

Value returned if logical test is **TRUE**

Value returned if logical test is **FALSE**

	A	B	C	D
1	Location	Temp (F)	Precip (mm)	Freeze
2	A	75	0	No
3	B	18	0	Yes
4	C	86	0	No
5	D	80	2.3	No
6	E	28	1.2	Yes
7	F	68	0.5	No
8	G	26	0	Yes

= IF(B2<=0,"Yes","No")

In this case we're categorizing the Freeze column as "Yes" if the temperature is equal to or below 32, otherwise "No"

Nested IF Statements

By using **Nested IF Statements**, you can include multiple logical tests within a single formula:

	A	B	C	D	E
1	Location	Temp (F)	Precip (mm)	Freeze	Climate
2	A	75	0	No	Mild
3	B	18	0	Yes	Cold
4	C	86	0	No	Hot
5	D	80	2.3	No	Mild
6	E	28	1.2	Yes	Cold
7	F	68	0.5	No	Mild
8	G	26	0	Yes	Cold

= IF(B2<40,"COLD",IF(B2>80,"HOT","MILD"))

If temp<40, climate = "Cold", if temp>80, climate = "Hot", otherwise climate = "Mild"

Excel's **AND** and **OR** statements allow you to include multiple logical tests at once:

	A	B	C	D	E	F	G
1	Location	Temp (F)	Precip (mm)	Freeze	Climate	Precip Type	Conditions
2	A	75	0	No	Mild	None	Dry
3	B	18	0	Yes	Cold	None	Dry
4	C	86	0	No	Hot	None	Dry
5	D	80	2.3	No	Mild	Rain	Wet
6	E	28	1.2	Yes	Cold	Snow	Wet
7	F	68	0.5	No	Mild	Rain	Wet
8	G	26	0	Yes	Cold	None	Dry

=IF(OR(F2="Rain",F2="Snow"),"Wet","Dry")

Here we're categorizing conditions as "Wet" if the precipitation type equals "rain" OR "snow", otherwise Conditions = "Dry"

=IF(AND(D2="Yes",C2>0),"Snow",IF(AND(D2="No",C2>0),"Rain","None"))

If the temp is below freezing AND the amount of precipitation > 0, then Precip Type = "Snow", if the temp is above freezing AND the amount of precipitation > 0, then Precip Type = "Rain", otherwise Precip Type = "None"



PRO TIP:

When writing nested functions, copy/paste repetitive pieces and tweak individual elements to save time (rather than starting from scratch)

If you want to evaluate a case where a logical statement is *not* true, you can use either the **NOT** statement or a "**<>**" operator

	A	B	C	D	E	F	G
1	Location	Temp (F)	Precip (mm)	Freeze	Climate	Precip Type	Conditions
2	A	75	0	No	Mild	None	Dry
3	B	18	0	Yes	Cold	None	Dry
4	C	86	0	No	Hot	None	Dry
5	D	80	2.3	No	Mild	Rain	Wet
6	E	28	1.2	Yes	Cold	Snow	Wet
7	F	68	0.5	No	Mild	Rain	Wet
8	G	26	0	Yes	Cold	None	Dry

=IF(NOT(C2=0),"Wet","Dry")

=IF(C2<>0,"Wet","Dry")

In both of these examples, we're defining Conditions = "Wet" if the amount of precipitation is NOT equal to 0

The **IFERROR** statement is an excellent tool to eliminate annoying error messages (**#N/A**, **#DIV/0!**, **#REF!**, etc.), which is particularly useful for front-end formatting

=IFERROR(value, value_if_error)

Formula or value that may or may not result in an error

Value returned in the case of an error



PRO TIP:

*If you're writing a formula that may trigger an error (i.e. a **VLOOKUP** where not all values have a match), **WRITE THE FULL FORMULA FIRST** then wrap it in an **IFERROR** statement*

Excel offers a number of different **IS** formulas, each of which checks whether a certain condition is true:

ISBLANK = Checks whether the reference cell or value is blank

ISNUMBER = Checks whether the reference cell or value is numerical

ISTEXT = Checks whether the reference cell or value is a text string

ISERROR = Checks whether the reference cell or value returns an error

ISEVEN = Checks whether the reference cell or value is even

ISODD = Checks whether the reference cell or value is odd

ISLOGICAL = Checks whether the reference cell or value is a logical operator

ISFORMULA = Checks whether the reference cell or value is a formula